

Cite this paper: *Chin. J. Chem.* **2022**, *40*, 1537–1545. DOI: 10.1002/cjoc.202200056

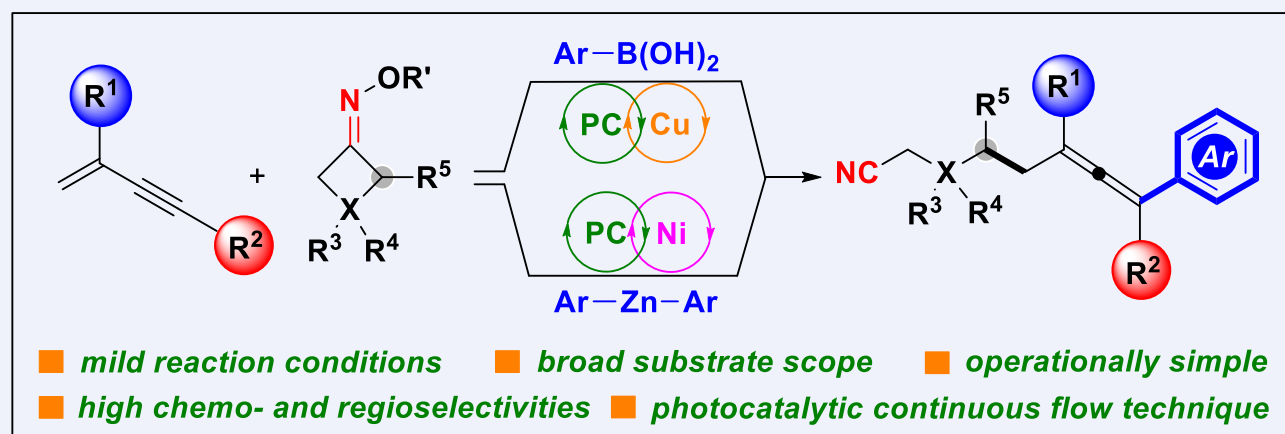
Photoinduced Merging with Copper- or Nickel-Catalyzed 1,4-Cyanoalkylarylation of 1,3-Enynes to Access Multiple Functionalized Allenes in Batch and Continuous Flow

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Comprehensive Summary

A three-component reaction of 1,3-enynes and cyclobutanone oxime esters in the presence of aryl boronic acids or organozinc reagents via the photoredox/copper or photoredox/nickel catalysis has been established. This redox-neutral 1,4-cyanoalkylarylation reaction has demonstrated mild condition, high catalytic reactivity and wide functional group compatibility, allowing access to a variety of functionalized tetra-substituted allene derivatives with high chemo- and regioselectivity. Moreover, using photocatalytic continuous flow technique to promote this process would result in increased yields (70% in flow vs. 61% in batch), reduced reaction times (7 min in flow vs. 6 h in batch), and easy scale-up (upgrade to gram scale), showcasing its potential as a synthetic platform.



Keywords

Photochemistry | 1,4-Cyanoalkylarylation | Radical reaction | 1,3-Enynes | 1,4-Cyanoalkylarylation | Tetra-substituted allenes | Photo catalytic continuous flow technique

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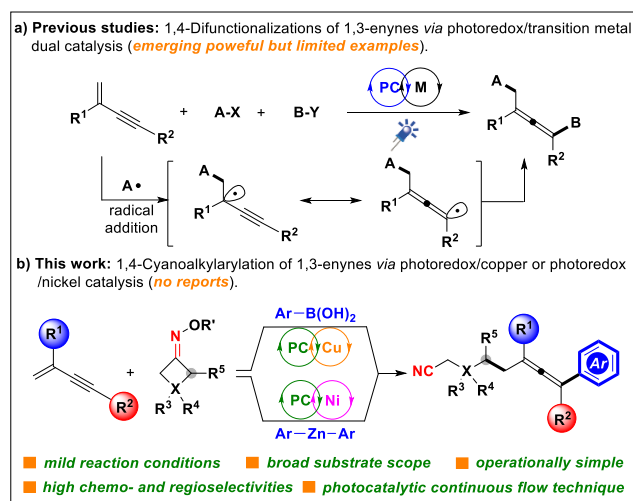
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Supporting Information

Background and Originality Content

Allene scaffolds are not only prevalent in active pharmaceuticals, agrochemicals and materials science, but also recognized as an essential class of synthetic intermediates for various organic synthesis.^[1] In this context, considerable attention has been focused on developing catalytic allene synthesis, and as a result, some elegant methods have been well established including molecular rearrangement,^[2] nucleophilic substitution,^[3] 1,4-addition^[4] and other methods.^[5] Compared to these classical approaches, the radical-mediated 1,4-difunctionalization of 1,3-enynes was considered as one of the most straightforward routes to introduce two functional groups simultaneously for the construction of multisubstituted allenes (Scheme 1a). Recently, the groups of Liu,^[6] Wang,^[7] Bao,^[8] Lu,^[9] and Ma^[10] have developed the methods for the 1,4-difunctionalization of 1,3-enynes *via* radical pathway. Despite such breakthroughs, there still exist several long-standing challenges in this field, *e.g.*, the low reactivity of the starting materials, the requirement of nucleophiles to terminate the reaction and the difficulty to control 1,2- or 1,4-radical addition products, which still limit their further development. Therefore, it is highly desirable to develop efficient methods for the radical-mediated 1,4-difunctionalization of 1,3-enynes under mild condition.

Scheme 1 1,4-Difunctionalization of 1,3-enynes *via* photoredox/transition metal dual catalysis



On the other hand, the merger of photoredox and transition metal catalysis, termed metallaphotoredox catalysis, has emerged as a powerful tool for achieving synthetically valuable transformations,^[11] among which, the most representative is the photoredox/copper^[12] and photoredox/nickel^[13] dual catalytic system. In this regard, metallaphotoredox catalysis has been developing at a great speed, finding remarkable applications in the 1,4-radical addition of 1,3-enynes. Very recently, the Lu and coworkers presented a photoredox/copper catalysis for the 1,4-carbocyanations of 1,3-enynes with alkyl *N*-hydroxyphthalimide esters and trimethylsilyl cyanide (TMSCN).^[9b] Later on, the Li's group developed a photoredox/copper catalytic 1,4-sulfonylarylation of 1,3-enynes using aryl iodides and sulfinate salts.^[14]

Despite impressive progress, there are still no reports on the 1,4-difunctionalization of 1,3-enynes for the synthesis of the same allene derivatives using both photoredox/copper and photoredox/nickel dual catalysis. On the basis of the above-mentioned studies of the 1,4-addition of 1,3-enynes in the metallaphotore-

dox catalytic system and recent study on the ring-opening of cyclobutanone oxime derivatives,^[15] we envisioned that the cyanoalkyl radicals produced by cyclobutanone oximes might initiate the 1,3-enyne substrates and produce allenyl radical intermediates, followed by copper or nickel-catalyzed cross-coupling with aryl boronic acids^[16] or organozinc reagents, providing a variety of tetrasubstituted allenes *via* different pathways. This protocol can efficiently facilitate the 1,4-difunctionalization of 1,3-enynes and access to cyanoalkylated allenes with excellent functional group tolerance and high chemo- and regioselectivity (Scheme 1b). Recently, continuous-flow photochemistry is gaining popularity in both academic and industrial settings due to many advantages. Compared with the traditional batch process, photochemical transformation in continuous flow reactor results in often faster, safer, and higher-yielding reactions due to factors such as easy amplification of effective energy transfer between light source and reaction vessel.^[17] We hypothesized that the application of continuous flow technique may improve the efficiency of this 1,4-cyanoalkylation process. If realized, these improvements will bring great hope for the application of this method in pharmaceutical industry and other fields. Herein, we wish to report a photoredox/copper and photoredox/nickel-catalyzed 1,4-cyanoalkylation reaction of 1,3-enynes by the use of photocatalytic continuous flow technique, providing a wide range of not easily accessible but useful tetrasubstituted allenes under mild conditions.

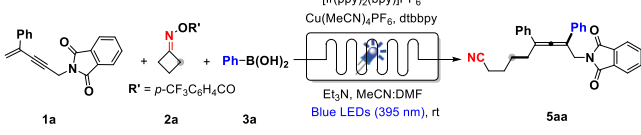
Results and Discussion

Our investigation began with the reaction of 1,3-enynes **1a** with cyclobutanone oxime ethers **2a** and phenyl boronic acid **3a** as the starting materials with the photoredox/copper dual catalytic system in a homemade continuous-flow photoreactor (details see Figure S1 in the Supporting Information). After extensive screening, the optimal conditions were achieved with 2.0 mol% PC-2 [Ir(ppy)₂(bpy)]PF₆ ($E_{1/2}^{IV/III} = -0.85$ V vs. SCE),^[18] 10.0 mol% Cu(MeCN)₄PF₆, 10.0 mol% 4,4'-di-*tert*-butyl-2,2'-bipyridine (dtbbpy, **L**₃), 2.0 equiv of Et₃N and 2.0 equiv of MeOH in MeCN:DMF co-solvent to irradiation with 395 nm purple LEDs with a residence time of 7 min, providing the desired tetrasubstituted allene **5aa** in 70% yield (see Tables S1–S8 in the Supporting Information). Slightly reduced reaction yields of **5aa** were observed when other photocatalysts (details see Table S5 in the Supporting Information), *e.g.*, fac-Ir(ppy)₃ ($E_{1/2}^{IV/III} = -1.73$ V vs. SCE)^[18] and [Ir(ppy)₂dtbbpy]PF₆ ($E_{1/2}^{IV/III} = -0.96$ V vs. SCE),^[18] were employed instead of PC-2 (Table 1, entries 2, 3). As for the choice of copper catalysts, CuI and Cu(MeCN)₄BF₄ demonstrated lower reactivity to that of Cu(MeCN)₄PF₆ (entries 4, 5). Screening the ligands revealed that **L**₃ was essential in order to realize this unique transformation. Lower yields were obtained when ligands **L**₁–**L**₂, **L**₄–**L**₅ took the place of **L**₃ as the ligand (entries 6, 7). However, use of other solvents, such as *N,N*-dimethylformamide (DMF), *N,N*-dimethylaniline (DMA), methanol (MeOH), failed to yield the **5aa** (entries 8, 9). The base triethylamine (Et₃N) was also necessary for this reaction. A series of other bases including 1,4-diazobicyclo-(2.2.2)octane (DABCO) and *N,N*-diisopropylethylamine (DIPEA), were also examined and found to result in lower yields (entries 10, 11). Finally, the reaction was conducted under the batch condition, with product **5aa** being isolated with 61% yield in 6 h (entry 12). Notably, the use of continuous-flow photoreactor could not only increase the reaction yield (70% in flow vs. 61% in batch), but also shorten the reaction time (7.0 min in flow vs. 6.0 h in batch).

With a set of optimized conditions in hand, we explored the scope of this reaction, with a first emphasis on the variations in the 1,3-enynes. To our delight, a wide array of functional groups with different electronic natures substituents (R¹, R²) at the 2- and

4-positions were well tolerated under optimal continuous flow conditions, providing various multi-functionalized allenes in moderate to good yields (more than 48 examples, Table 2).

Table 1 Optimization of the conditions^a



Entry	Deviation from standard conditions ^a	Yield ^b /%
1	None	70
2	PC-1 instead of PC-2	42
3	PC-5 instead of PC-2	47
4	CuI instead of Cu(MeCN) ₄ PF ₆	25
5	Cu(MeCN) ₄ BF ₄ instead of Cu(MeCN) ₄ PF ₆	52
6	L ₁ , L ₂ instead of L ₃	46–50
7	L ₄ –L ₆ instead of L ₃	31–53
8	DMF, DMA, MeOH instead of MeCN:DMF	Trace
9	DCM:DMF, THF:DMF instead of MeCN:DMF	22–29
10	DABCO, TMEDA, DMAP instead of Et ₃ N	Trace
11	DIPEA, DBU, Et ₂ NH instead of Et ₃ N	47–53
12	In batch	61 ^c

^a Reaction conditions: Feed I contains **1a** (0.2 mmol, 1.0 eq.) and **2a** (0.4 mmol, 2.0 eq.) and [Ir(ppy)₂(bpy)]PF₆ (2.0 mol%), Cu(MeCN)₄PF₆ (10.0 mol%), dtbbpy (10.0 mol%) and 16 μ L of MeOH (0.4 mmol, 2.0 eq.) in 1.5 mL of the solvent (MeCN:DMF = 2:1). Feed II contains **3a** (0.6 mmol, 3.0 eq.) and 56 μ L of Et₃N (0.4 mmol, 2.0 eq.) in 1.5 mL of the solvent (MeCN:DMF = 2:1). Irradiated with light from 20 W purple LEDs (λ = 395 nm) at room temperature. Residence time of 7.0 min. ^b Reported yields were based on **1a**. ^c Standard conditions: **1a** (0.2 mmol, 1.0 eq.) and **2a** (0.4 mmol, 2.0 eq.) and **3a** (0.6 mmol, 3.0 eq.), [Ir(ppy)₂(bpy)]PF₆ (2.0 mol%), Cu(MeCN)₄PF₆ (10.0 mol%), dtbbpy (10.0 mol%), 56 μ L of Et₃N (0.4 mmol, 2.0 eq.) and 16 μ L of MeOH (0.4 mmol, 2.0 eq.) in 3.0 mL of the solvent (MeCN:DMF = 2:1), irradiated with light from 20 W purple LEDs (395 nm) for 6 h at room temperature.

When the C2-position (R¹) was phenyl substituted, different alkyl phthalimide protecting groups located at the C4-position (R²) presented no obvious effects on this transformation, giving the corresponding cyanoalkylarylated allenes **5aa**–**5ae** with yields ranging from 53% to 70%. Other alkyl amine substituted functional groups, such as NHBz (**1f**) and NHTs (**1g**) located at C4-position (R²) were also tolerated, leading to the desired products **5af**, **5ag** in 61% and 62% yields, respectively. Additionally, the alcohol groups (OBn, OTBS) substituted on R² were also compatible for this reaction, giving the target products **5ah** and **5ai** in yields of 45% and 50%, respectively. In addition, an array of phenyl (**1j**–**1m**) and alkyl (**1n**, **1o**) enyne substrates were all generated smoothly, giving the corresponding allene products **5aj**–**5ao** in 43%–55% yields. Similarly, for the C2-methyl substituent of the 1,3-enynes, a variety of functional groups, including aryl (**1p**–**1v**), heteroaryl (**1w**), and amide (**1x**, **1y**) groups, were all well accommodated, to afford the corresponding products **5ap**–**5ay** in 35%–61% yields. It should be noted that when R¹ was H-substituted, it had little effect on this reaction, delivering the product **5az** in 61% yield.

Next, the scope of aryl boronic acids was evaluated. As expected, a series of aryl boronic acids bearing substituents includ-

ing Me, Cl, Br, and OBn groups, located on the *ortho*-, *meta*- as well as *para*-substituents of the phenyl ring were also suitable for this transformation, affording the target products **5ba**–**5bg** in 50%–71% yields. Encouraged by these results, we then turned our attention to examination of the scope of cyclobutanone O-benzoyl oximes. Notably, the 3- or 4-monosubstituted cyclobutanone O-benzoyl oximes bearing Me, Et, Bn, Ph, 4-ClPh, CO₂tBu and OBn groups underwent this radical cyanoalkylarylation reaction efficiently to give the products **5ca**–**5cg** in 40%–53% yields. Moreover, 3,3-disubstituted cyclobutanone derivatives were well-tolerated, and gave the expected products **5ch**, **5ci** in 61% and 72% yields, respectively. In addition, heterocyclobutan-3-one oxime esters (O) oximes were also compatible with this reaction, generating the tetrasubstituted allene **5cj** in 50% yield (Table 3).

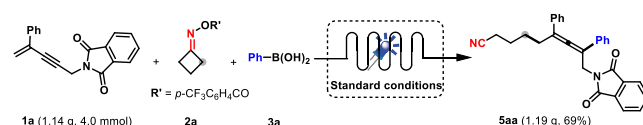
Currently, the photoredox/nickel-catalyzed reactions used to perform multicomponent difunctionalization of unsaturated hydrocarbon are of vital importance in modern organic synthesis.^[19] However, there are still few reports of the photoredox/nickel-catalyzed 1,4-functionalization of 1,3-enynes using organozinc reagents.^[7] For this reason, we attempted to carry out the reaction using the 1,3-enynes **1a**, cyclobutanone O-benzoyl oximes **2a**, and Ph₂Zn **4a** as the substrates with the photoredox/nickel dual catalytic systems (See Figure S3 in the Supporting Information). The use of organozinc reagents was easy to make the mixer and microreactor clogging, so we chose to conduct this reaction under the batch conditions.

Subsequently, we focused on screening the aryl zinc reagents to furnish adducts **5ab'**–**5af'** (5 examples). To our delight, a range of organozinc reagents bearing –Me, –Cl, –Ph, and –OMe substituents at the *para*-position of the aromatic ring successfully furnished the desired allene products **5ab'**–**5ae'** in 67%–80% yields. The dimethoxy substituted on aromatic ring of the organozinc reagent also worked well, and furnished the product **5af'** in 50% yield.

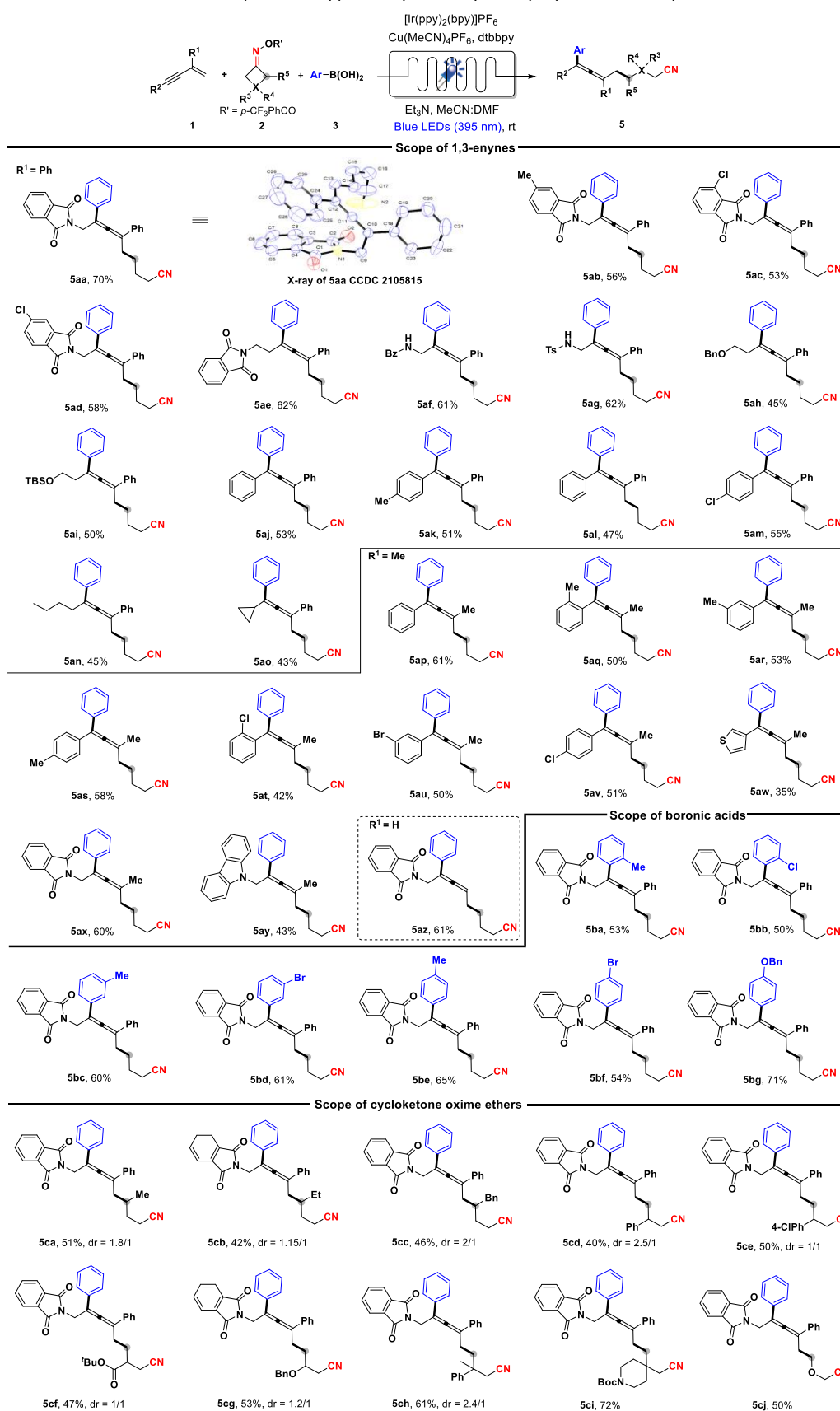
To further explore the utility of our procedure, we focused our attention on performing photoredox/copper dual catalytic reaction on a scale of 4.0 mmol (1.14 g) using the continuous flow technique to enhance the rate and progress of the reaction (Scheme 2). We chose to set up a continuous flow procedure in a homemade photo-flow reactor that ensured convenience, reproducibility, and scalability. To our delight, the desired product **5aa** was obtained in the yield of 69%. These results further highlighted the comprehensive potential of our current methods.

To gain some insight into the 1,4-cyanoalkylarylation reaction, we initially carried out several control experiments with the model substrates **1a**, **2a** and **3a**. Firstly, the addition of 2.0 equivalents of 2,2,6,6-tetramethyl-1-piperidinyloxy (TEMPO) or diphenyl diselenide (PhSeSePh) inhibited the formation of the desired product **5aa**, providing the radical-trapping products **6** and **7** detected by high-resolution mass spectrometry or nuclear magnetic resonance analysis (Scheme 3a).

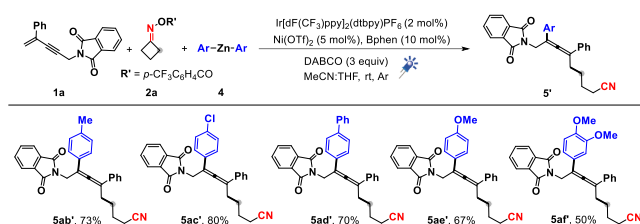
Scheme 2 Late-stage functionalization using the continuous flow technique^a



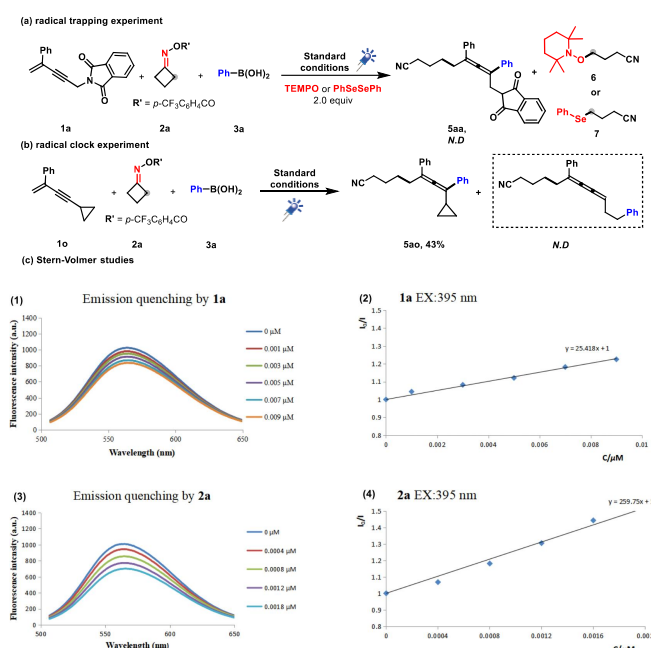
^a Reaction conditions: Feed I contains **1a** (4.0 mmol, 1.0 eq.) and **2a** (8.0 mmol, 2.0 eq.) and [Ir(ppy)₂(bpy)]PF₆ (2.0 mol%), Cu(MeCN)₄PF₆ (10.0 mol%), dtbbpy (10.0 mol%) and 320 μ L of MeOH (8.0 mmol, 2.0 eq.) in 15 mL of the solvent (MeCN:DMF = 2:1). Feed II contains **3a** (12.0 mmol, 3.0 eq.) and 1.12 mL of Et₃N (8.0 mmol, 2.0 eq.) in 15 mL of the solvent (MeCN:DMF = 2:1). Irradiated with light from 20 W purple LEDs (λ = 395 nm) at room temperature. Residence time of 7.0 min. ^b Reported yields were based on **1a**.

Table 2 Scope of the copper-catalyzed 1,4-cyanoalkylarylation of 1,3-enynes^a

^a Reaction conditions: Feed I contains **1a** (0.2 mmol, 1.0 eq.) and **2a** (0.4 mmol, 2.0 eq.) and [Ir(ppy)₂(bpy)]PF₆ (2.0 mol%), Cu(MeCN)₄PF₆ (10.0 mol%), dtbbpy (10.0 mol%) and 16 μL of MeOH (0.4 mmol, 2.0 eq.) in 1.5 mL of the solvent (MeCN : DMF = 2 : 1). Feed II contains **3a** (0.6 mmol, 3.0 eq.) and 56 μL of Et₃N (0.4 mmol, 2.0 eq.) in 1.5 mL of the solvent (MeCN : DMF = 2 : 1). Irradiated with light from 20 W purple LEDs (λ = 395 nm) at room temperature. Residence time of 7.0 min. ^b Reported yields were based on **1a**.

Table 3 Scope of the nickel-catalyzed 1,4-cyanoalkylarylation of 1,3-enynes^a

^a Standard conditions: **1a** (0.2 mmol, 1.0 eq.), **2a** (0.3 mmol, 1.5 eq.), **4** (0.2 mmol, 1.0 eq.), Ir[dF(CF₃)ppy]₂(dtbbpy)PF₆ (2.0 mol%), Ni(OTf)₂ (5.0 mol%), Bphen (10.0 mol%), DABCO (0.6 mmol, 3.0 eq.), solvent 3.0 mL (DMF : THF = 1 : 2), irradiated with light from 20 W blue LEDs (λ = 455 nm) at room temperature for about 12 h. ^b Isolated yield of each recrystallized product was calculated based on the yield of **1a**.

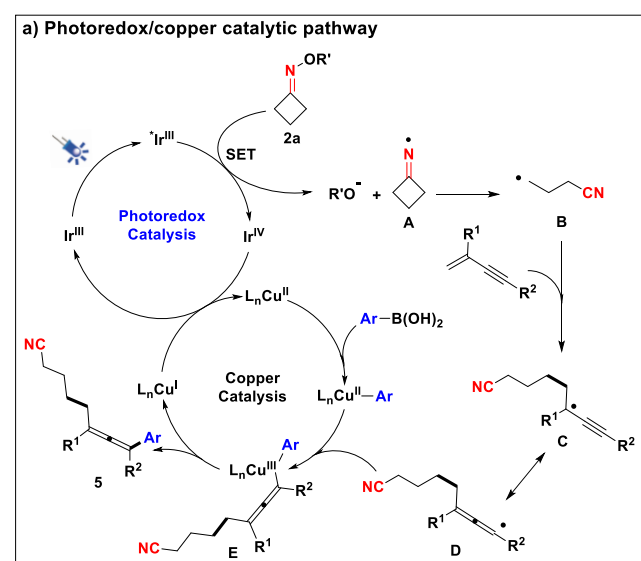
Scheme 3 Mechanistic experiments

Mechanistic studies: (a) radical trapping experiment; (b) radical clock experiment; (c) fluorescence quenching experiments with **PC-2** and various concentrations of **1a** and **2a**.

Both of these results suggested that the cyanoalkyl radical was involved in the process. The radical clock experiments were performed to further confirm this speculation. The radical clock **1o** was added to the system instead of conventional 1,3-enynes, yielding the no ring-opening product **5ao** in 43% yield (Scheme 3b), which showed that the coupling of the allenyl radical to the Cu(II) complex was much faster. Finally, we conducted the Stern-Volmer studies, and the results indicated that the excited-state ^{*}Ir^{III} could be quenched by the cyclobutanone *O*-benzoyl oximes more efficiently than by 1,3-enynes, which indicated a reductive quenching mechanism (Scheme 3c).

On the basis of the above mechanistic studies and literature precedents,^[2-17] we propose the photoredox/copper catalytic cycles involve a possible cyanoalkyl radical mechanism in Scheme 4. Initially, the active Ir^{III} species is triggered by the visible light induced and produces the excited photocatalyst ^{*}Ir^{III} complex. A single electron transfer from the excited states of ^{*}Ir^{III} to the *O*-acyl oxime **2a** results in the generation of the iminyl radical **A**, an alkoxy anion, and an Ir^{IV} complex. Then, iminyl radical **A** undergoes a ring-opening by homolytic C—C bond fragmentation to

form a highly reactive cyanoalkyl radical **B**, which can be intercepted by the 1,3-enyne **1** followed by radical shift to give the allenyl radical species **D**. Alternatively, the Cu(II) species could form an Ar-Cu(II) species by exchanging its ligand with ArB(OH)₂. Next, the Ar-Cu(II)L_n species is oxidized by the allenyl radical species **D** to form the Cu(III) intermediate, which undergoes reductive elimination to provide the target tetra-substituted allene **5** and the Cu(I) species. Afterwards, single electron oxidation of the Cu(I) species by Ir^{IV} complex generates the Cu(II) species with the concomitant regeneration of the photocatalyst Ir^{III}, closing the photoredox cycle. Similarly, the photoredox/nickel catalytic cycle was similar to the photoredox/copper catalytic cycle (For details see Figure S5 in the Supporting Information).

Scheme 4 Proposed mechanism

Conclusions

We have developed the first example of the 1,4-difunctionalization of 1,3-enynes for the synthesis of a range of tetra-substituted allenes *via* the photoredox/copper or photoredox/nickel dual catalysis. An array of synthetically useful cyanoalkyl- and aryl-containing tetrasubstituted allenes were obtained in good yields, with excellent chemoselectivities as well as remarkable functional group compatibility under mild reaction conditions. Furthermore, this strategy has also proved to be efficient, with the potential for scalability by the use of the continuous-flow technique. We believe this method offers not only a straightforward approach for the convenient and efficient construction of allene derivatives but previously inaccessible structures that were found as structural motifs in various natural products or drug molecules.

Experimental

General procedure for copper-catalyzed 1,4-cyanoalkylarylation of 1,3-enynes

1,3-Enyne **1a** (0.2 mmol, 1.0 eq.), cyclobutanone oxime esters **2a** (0.4 mmol, 2.0 eq.), [Ir(ppy)₂(bpy)]PF₆ (2.0 mol%), Cu(MeCN)₄PF₆ (10.0 mol%), dtbbpy (10.0 mol%) and 16 μ L of MeOH (0.4 mmol, 2.0 eq.) were dissolved in 1.5 mL of the solvent (MeCN : DMF = 2 : 1) as Feed I. **3a** (0.6 mmol, 3.0 eq.) and 56 μ L of Et₃N (0.4 mmol, 2.0 eq.) were dissolved in 1.5 mL of the solvent (MeCN : DMF = 2 : 1) as Feed II. Then the two feeds were trans-

ferred into two 5 mL BD plastic syringes and introduced into the microreactor (a high purity perfluoroalkoxyalkane, PFA capillary tubing, ID = 600 μ m) through syringe pump. The two liquid streams were merged with a Y-Mixer before entering the microreactor. The flow rate was set to 0.2 mL/min (0.1 mL/min per syringe), thus resulting in 7.0 min residence time (volume of reactor = 1.4 mL). 20 W purple LEDs (395 nm) was used as light source. After reaching steady state, the reaction sample was collected in a glass vial. Solution remaining in the microreactor was then discharged with MeCN (2.0 mL $\times$ 2) via syringe pump, and was also collected in the same glass vial. The collected reaction solution was extracted with EtOAc (3 $\times$ 25 mL) and brine, and the organic layer was collected, dried with Na₂SO₄ and evaporated under reduced pressure. The resulting crude compound was absorbed on silica gel and purified via column chromatography (EtOAc/PE). The isolated compound was analyzed by HRMS and NMR.

General procedure for nickel-catalyzed 1,4-cyanoalkylarylation of 1,3-enynes

Under argon atmosphere, an oven-dried resealable Schlenk tube equipped with a magnetic stir bar was charged with 1,3-enyne (**1a**, 0.2 mmol, 1.0 eq.), cyclobutanone oxime ester (**2a**, 0.3 mmol, 1.5 eq.), Ni(OTf)₂ (5.0 mol%), Bphen (10.0 mol%), Ir[d(CF₃)ppy]₂(dtbbpy)PF₆ (2.0 mol%), DABCO (0.6 mmol, 3.0 eq.), anhydrous THF (1.4 mL), and anhydrous DMF (1.0 mL). Then, Ph₂Zn (**4a**, 0.2 mmol, 1.0 eq., c = 0.33 mol/L in THF) was added into the mixture. After that, the solution was stirred at a distance of ~5 cm from a 20 W blue LEDs (455 nm) at room temperature for about 12 h. The reaction solution was extracted with EtOAc (3 $\times$ 25 mL) and chloridamonyl, and the organic layer was collected, dried with Na₂SO₄ and evaporated under reduced pressure. The resulting crude compound was absorbed on silica gel and purified via column chromatography (EtOAc/PE, varying ratios). The isolated compound was analyzed by HRMS and NMR.

Supporting Information

The supporting information for this article is available on the WWW under <https://doi.org/10.1002/cjoc.202200056>.

Acknowledgement

This work was supported by the National Natural Science Foundation of China (Nos. 21702103, 21522604), the Jiangsu Synergetic Innovation Center for Advanced Bio-Manufacture (No. XTD2203), and the Natural Science Research Projects of Jiangsu Higher Education (No. 19KJB150027).

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Manuscript received: February 9, 2022

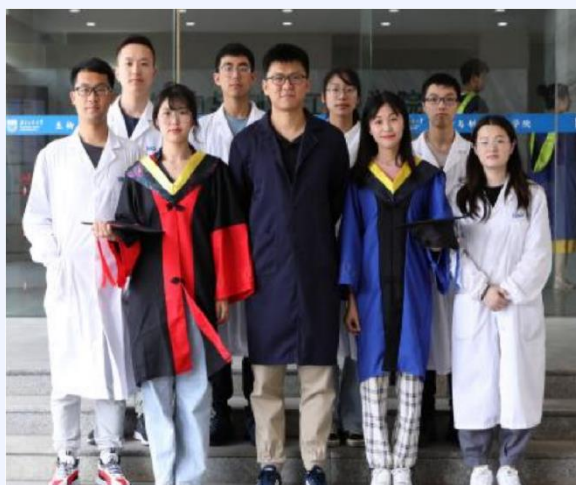
Manuscript revised: March 22, 2022

Manuscript accepted: March 27, 2022

Accepted manuscript online: March 31, 2022

Version of record online: April 20, 2022

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